

Extended Vehicle Speed Control System - Extended High-Level Design

Document ID: HLD-VSC-100-001 Version: 1.0 Status: Synthetic Demonstration Document

1. Purpose and Scope

This synthetic AUTOSAR-style HLD describes the Extended Vehicle Speed Control System. It is intentionally long to test page-wise ingestion, chunking, semantic retrieval, structured architecture extraction, validation, graph construction, revision tooling, and evidence-grounded question answering. The architecture facts are deterministic; surrounding engineering prose provides realistic document length without introducing confidential OEM data.

2. Architecture Summary

Component	Responsibility	Key Interface
WheelSpeedSensor	Acquires wheel-speed information and publishes it to vehicle-control functions	WheelSpeedInterface
VehicleSpeedController	Processes wheel-speed information and computes vehicle speed.	VehicleSpeedInterface
InstrumentCluster	Displays vehicle speed to the driver.	DisplaySpeedInterface
DiagnosticsManager	Monitors vehicle-speed processing and publishes diagnostic state.	DiagnosticStatusInterface
SpeedWarningManager	Evaluates configurable speed thresholds and publishes warnings.	SpeedWarningInterface
DataLogger	Records selected vehicle-speed and diagnostic information.	LoggingInterface
GatewayAdapter	Exposes selected vehicle-speed information to other vehicle domains.	GatewaySpeedInterface
HealthSupervisor	Monitors health status of vehicle-speed related components.	SystemHealthInterface

2.1 Document Processing Guidance

The document should be processed incrementally by page. Native text extraction is preferred; OCR is a fallback for image-only pages. Retrieval should operate on chunks rather than passing the complete document to the language model. Architecture extraction should normalize equivalent labels before mapping facts into a common architecture model.

3. Component Definitions

Component Name	Description
WheelSpeedSensor	Acquires wheel-speed information and publishes it to vehicle-control functions.
VehicleSpeedController	Processes wheel-speed information and computes vehicle speed.
InstrumentCluster	Displays vehicle speed to the driver.
DiagnosticsManager	Monitors vehicle-speed processing and publishes diagnostic state.
SpeedWarningManager	Evaluates configurable speed thresholds and publishes warnings.
DataLogger	Records selected vehicle-speed and diagnostic information.
GatewayAdapter	Exposes selected vehicle-speed information to other vehicle domains.
HealthSupervisor	Monitors health status of vehicle-speed related components.

4. Interface Definitions

Interface Name	Provider	Consumer
WheelSpeedInterface	WheelSpeedSensor	VehicleSpeedController
VehicleSpeedInterface	VehicleSpeedController	InstrumentCluster
DisplaySpeedInterface	InstrumentCluster	DataLogger
DiagnosticStatusInterface	DiagnosticsManager	HealthSupervisor
SpeedWarningInterface	SpeedWarningManager	InstrumentCluster
LoggingInterface	DataLogger	GatewayAdapter
GatewaySpeedInterface	GatewayAdapter	ExternalDomain
SystemHealthInterface	HealthSupervisor	DiagnosticsManager

5. Port Definitions

Component	Port Name	Port Type	Interface
WheelSpeedSensor	WheelSpeedSensorOut	P-Port	WheelSpeedInterface
VehicleSpeedController	VehicleSpeedControllerOut	P-Port	VehicleSpeedInterface
InstrumentCluster	InstrumentClusterOut	P-Port	DisplaySpeedInterface
DiagnosticsManager	DiagnosticsManagerOut	P-Port	DiagnosticStatusInterface
SpeedWarningManager	SpeedWarningManagerOut	P-Port	SpeedWarningInterface
DataLogger	DataLoggerOut	P-Port	LoggingInterface
GatewayAdapter	GatewayAdapterOut	P-Port	GatewaySpeedInterface
HealthSupervisor	HealthSupervisorOut	P-Port	SystemHealthInterface
VehicleSpeedController	WheelSpeedIn	R-Port	WheelSpeedInterface
InstrumentCluster	VehicleSpeedIn	R-Port	VehicleSpeedInterface
HealthSupervisor	DiagnosticStatusIn	R-Port	DiagnosticStatusInterface
InstrumentCluster	SpeedWarningIn	R-Port	SpeedWarningInterface
GatewayAdapter	LoggingIn	R-Port	LoggingInterface
DiagnosticsManager	SystemHealthIn	R-Port	SystemHealthInterface

6. Signal Definitions

Signal Name	Interface	Unit
WheelSpeed	WheelSpeedInterface	km/h
VehicleSpeed	VehicleSpeedInterface	km/h
DiagnosticStatus	DiagnosticStatusInterface	enum
SpeedWarning	SpeedWarningInterface	boolean
SystemHealth	SystemHealthInterface	enum

7. Component Dependencies

Dependent Component	Required Component
VehicleSpeedController	WheelSpeedSensor
InstrumentCluster	VehicleSpeedController
DiagnosticsManager	VehicleSpeedController
SpeedWarningManager	VehicleSpeedController
DataLogger	VehicleSpeedController
DataLogger	DiagnosticsManager
GatewayAdapter	DataLogger
HealthSupervisor	DiagnosticsManager

8. Functional Flows

Flow ID	Architecture Flow
FLOW-01	WheelSpeedSensor -> VehicleSpeedController -> InstrumentCluster
FLOW-02	VehicleSpeedController -> SpeedWarningManager -> InstrumentCluster
FLOW-03	VehicleSpeedController -> DiagnosticsManager -> HealthSupervisor
FLOW-04	VehicleSpeedController -> DataLogger -> GatewayAdapter

8.1 Traceability Notes

Each flow is derived from explicit component, interface, port, signal, and dependency relationships. Generated analysis must remain traceable to the source pages and must not be treated as automatic architecture approval.

6. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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7. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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8. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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9. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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10. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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11. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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12. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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13. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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14. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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15. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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16. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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17. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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18. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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19. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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20. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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21. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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22. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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23. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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24. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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25. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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26. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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27. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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28. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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29. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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30. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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31. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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32. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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33. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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34. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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35. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

This page intentionally contains enough technical prose to create multiple retrieval chunks across the full 100-page document. A scalable implementation should index chunks once, retrieve only the most relevant evidence for each question, and avoid sending all pages to the language model. Page metadata should be retained so every answer can be traced back to its source.

36. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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37. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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38. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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39. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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40. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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41. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

This page intentionally contains enough technical prose to create multiple retrieval chunks across the full 100-page document. A scalable implementation should index chunks once, retrieve only the most relevant evidence for each question, and avoid sending all pages to the language model. Page metadata should be retained so every answer can be traced back to its source.

42. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

This page intentionally contains enough technical prose to create multiple retrieval chunks across the full 100-page document. A scalable implementation should index chunks once, retrieve only the most relevant evidence for each question, and avoid sending all pages to the language model. Page metadata should be retained so every answer can be traced back to its source.

43. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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44. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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45. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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46. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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47. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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48. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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49. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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50. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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51. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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52. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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53. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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54. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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55. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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56. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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57. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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58. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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59. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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60. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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61. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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62. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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63. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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64. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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65. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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66. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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67. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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68. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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69. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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70. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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71. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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72. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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73. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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74. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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75. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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76. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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77. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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78. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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79. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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80. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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81. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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82. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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83. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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84. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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85. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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86. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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87. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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88. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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89. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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90. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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91. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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92. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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93. Engineering Detail - HealthSupervisor

Responsibility. Monitors health status of vehicle-speed related components.

Primary interface. SystemHealthInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

HealthSupervisor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace HealthSupervisor to SystemHealthInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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94. Engineering Detail - WheelSpeedSensor

Responsibility. Acquires wheel-speed information and publishes it to vehicle-control functions.

Primary interface. WheelSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

WheelSpeedSensor participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace WheelSpeedSensor to WheelSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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95. Engineering Detail - VehicleSpeedController

Responsibility. Processes wheel-speed information and computes vehicle speed.

Primary interface. VehicleSpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

VehicleSpeedController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace VehicleSpeedController to VehicleSpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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96. Engineering Detail - InstrumentCluster

Responsibility. Displays vehicle speed to the driver.

Primary interface. DisplaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InstrumentCluster participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InstrumentCluster to DisplaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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97. Engineering Detail - DiagnosticsManager

Responsibility. Monitors vehicle-speed processing and publishes diagnostic state.

Primary interface. DiagnosticStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DiagnosticsManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DiagnosticsManager to DiagnosticStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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98. Engineering Detail - SpeedWarningManager

Responsibility. Evaluates configurable speed thresholds and publishes warnings.

Primary interface. SpeedWarningInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

SpeedWarningManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace SpeedWarningManager to SpeedWarningInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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99. Engineering Detail - DataLogger

Responsibility. Records selected vehicle-speed and diagnostic information.

Primary interface. LoggingInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DataLogger participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DataLogger to LoggingInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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100. Engineering Detail - GatewayAdapter

Responsibility. Exposes selected vehicle-speed information to other vehicle domains.

Primary interface. GatewaySpeedInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

GatewayAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace GatewayAdapter to GatewaySpeedInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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